Team Atlas, Rutgers University-NB
2026 M.A.J.I.C. Challenge

Anomaly Detection at the Thailand-Cambodia Border Using Attention-Based LSTMs

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Problem and Approach

Problem

- Conflict is unpredictable.
- Sudden escalation after periods of calm.
- Difficult to forecast using traditional methods.



Approach

- Build an early warning system.
- Combine conflict data and media signals.
- Detect unusual patterns over time.
- Outputs a risk signal, not a direct prediction.

Background of Conflict

Timeline

- Pre 1965
- 2001
- 2008-2011
- 2025-Present

Geographical

- Koh Kood and Gulf areas
- Preah Vihear
- Ta Muen Thom

Political

- Media coverage
- Election years
- Leadership change
- Diplomatic intervention

Economical

- Resource demarcation
- Supply and demand
- Int. trade

Outlook

Pattern:

- Low intensity but recurring.
- Long periods of calm followed by escalation.

Challenge:

- Politically sensitive and difficult to predict.

Open-Source Datasets We Used

1. Armed Conflict Location and Event Data Project (ACLED)

- Semi-geolocated conflict points
- Define variable *Violence* = sum of events per day



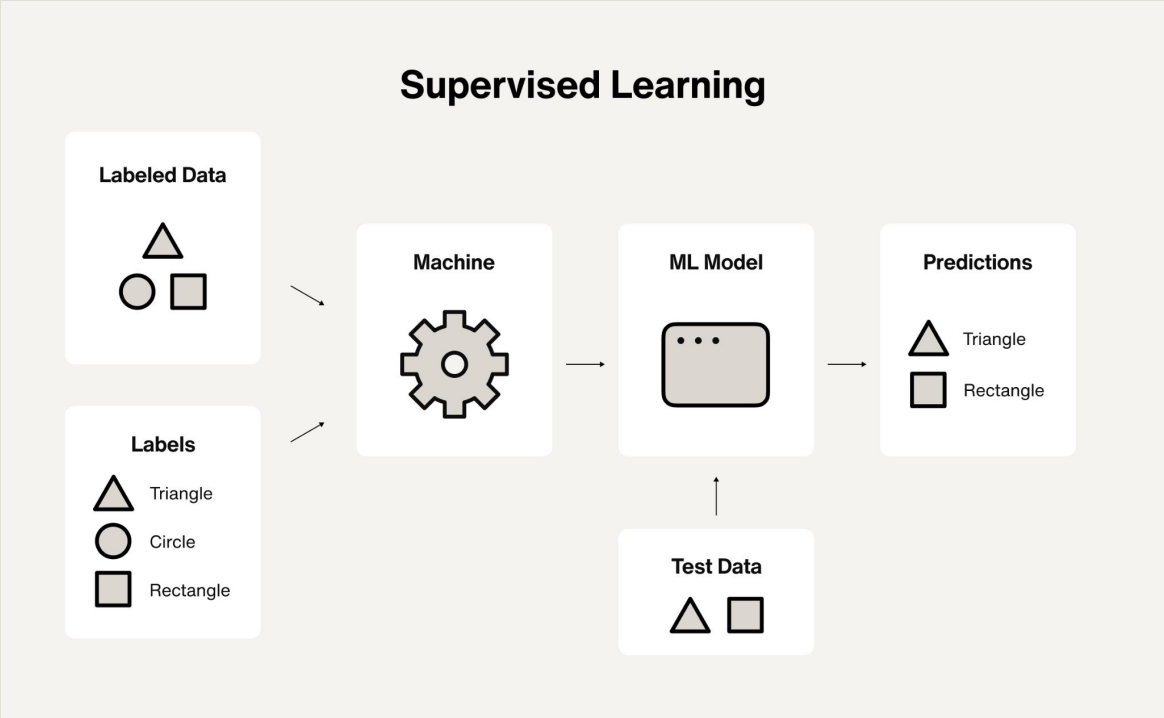
GDELT 2.0

2. Global Database of Events, Language, and Tone (GDELT)

- Semi-geolocated global news media
- Custom scoring based on article count, impact rating, and tone
- Discard non-geolocated data

Combine these into PRIO-GRID -> monitor violence in each cell!

Background: Traditional Supervised Learning



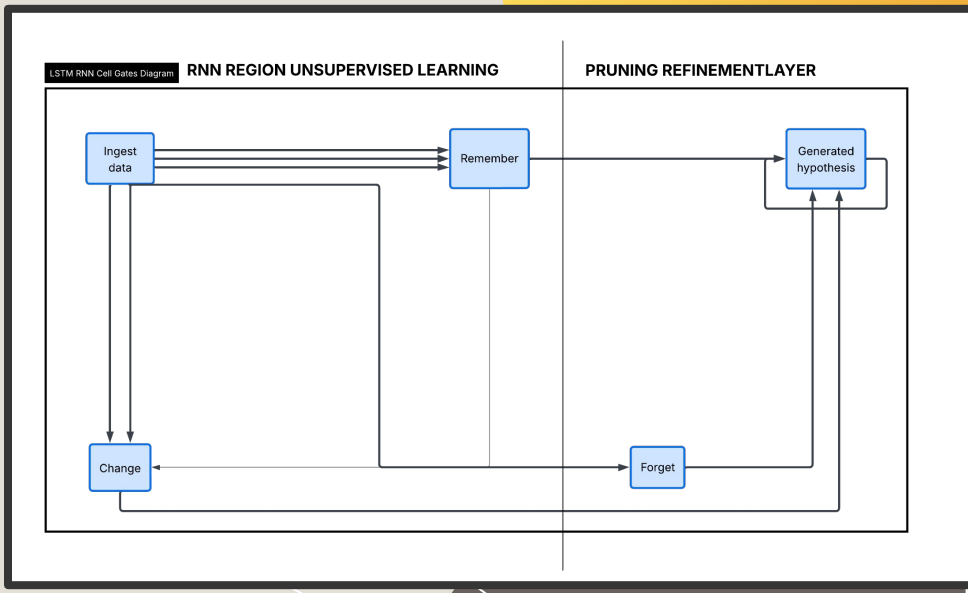
Model Design

Concept

- Learns to reconstruct “normal activity”
- As conflict intensifies, reconstruction error increases

Architecture

- Long short-term memory (LSTM) captures temporal patterns
- Autoencoder reconstructs normal behavior
- Attention highlights spatial correlation and important signals from history



LSTM Cell

Results

Key Findings:

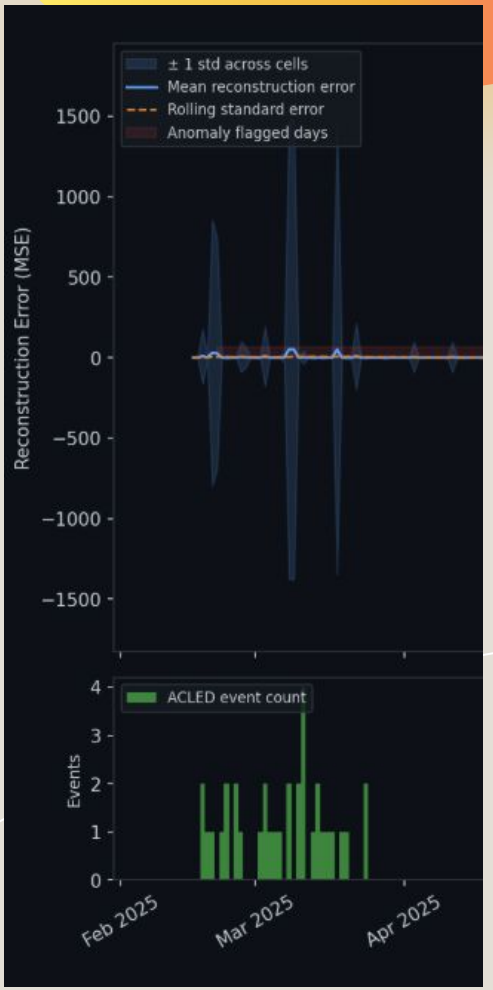
- Model predicts mostly non-violence.
- Error increases during escalation periods.

Insights:

- Reconstruction error signals rising risk.

Outcome:

- Enables early detection of conflict escalation.



Impact and Limitations

Impact:

- Supports early warning systems.
- Informs policy and humanitarian response.
- Informs monitoring of conflict zones.

Limitations:

- Data delays (ACLED, GDELT)
- Conflict is inherently unpredictable.
- Model signals escalation, not direct causes.

Future Studies

- ❑ Geospatial Integration
- ❑ Real-Time Data Sources
- ❑ Data Pipeline Improvements
- ❑ Advanced Spatial Modeling
- ❑ Hybrid Modeling Approach
- ❑ Political Context Integration

Thank You!

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